



Open Mathematical Olympiad for
University Students (OMOUS-2026)
Ashgabat, 23-28 March



Language: English

Tuesday, 24-March

INDIVIDUAL COMPETITION
Category A

Problem 1

During the year of "Independent neutral Turkmenistan – the homeland of purposeful winged horses" total 52 horse races were held. At each one 5 best horses were awarded. The horse that was awarded at every horse race of that year called *champion*. It is known that the prize receivers of every two races had exactly 1 horse in common. Show that there exists a *champion* horse who got a prize at every horse race of that year.

Problem 2

Let $\{a_n\}$ be a sequence of numbers such that $a_0 = 1, a_1 = 2, a_{n+1} = \frac{3^{n-1} + a_n^2}{a_{n-1}}$ for $n \geq 1$.

- Show that all a_n are integers.
- Does the series $\sum_{n=0}^{\infty} \frac{1}{a_n}$ converge?
- Does the series $\sum_{n=0}^{\infty} \frac{a_{n+1}}{a_n}$ converge?

Problem 3

A chord is drawn in a conic section such that at one of its endpoints, it is perpendicular to the tangent of the curve. Find the direction of the chord (the angle formed with the axis of the conic section) for which the area of the segmented portion is minimal, and express this area in terms of the parameters of the conic section.

Problem 4

Let n be a positive integer and let $A, B \in M_n(\mathbb{C})$ such that $B^2 = B$ and $A^2 = A$. Prove that

$$\text{rank}(A + B - AB) \geq \text{rank}(A - B).$$

Problem 5

Is there a positive integer n such that for each nonzero digit, the numbers $n, 2n, 3n, \dots, 62026n$ have the same digits up to multiplicity of each nonzero digit?

Problem 6

Let f be a real-valued function on $[0, 1]$ with continuous and non-negative fifth derivative such that $\int_0^1 f(x) dx = \int_0^1 x^5 f(x) dx = 0$. Prove that

$$\int_0^1 (56x^3 + 3x) f(x) dx \geq \int_0^1 (63x^4 + 21x^2) f(x) dx.$$

Find all the functions that give equality in the above inequality.

$$\begin{aligned} & \frac{1}{4} \int_0^1 \left(\binom{8}{5} x^3 + \frac{1}{2} \binom{6}{5} x \right) f(x) dx \geq \\ & \geq \int_0^1 \left(\frac{1}{2} \binom{9}{5} x^4 + \binom{7}{5} x^2 \right) f(x) dx \end{aligned}$$

Time: 4 hours 30 minutes.
Each problem is worth 10 points.

Problem 1

During the year 52 horseraces were held. At each one 5 best horses were awarded. It is known that the prize receivers of every two races had exactly 1 horse in common. Show that there exists a horse who got a prize at every horse race.

Solution:

Look at an arbitrary race, let that be A_1 , where the prizes went to some 5 horses. Each of the remaining 51 races had to have some horse among those 5 receiving a prize. By pigeonhole principle there exists a horse that in addition to A_1 also got a prize at least 10 races. Let that horse be a and those races be $A_2, \dots, A_{11}$. Let now B be an arbitrary race that is different from $A_1, \dots, A_{11}$. As each one of the races $A_1, \dots, A_{11}$ has one prize-winning horse in common with B and exactly 5 horses get prizes at B , applying pigeonhole principle again shows that one of those five had to get a prize at least three of $A_1, \dots, A_{11}$. Since these three have horse a in common and according to initial conditions that horse is the only one, this means that a also got a prize at race B . But since we picked B arbitrarily, a must have got a prize at every race.

Marking scheme Problem 1

1.	Applying Pigeonhole principal without (2)	1 point
2.	Proving existence of one winner for $A_1, A_2, \dots, A_{11}$	+2 point
3.	Claiming for any B that the common horse of $A_1, A_2, \dots, A_{11}$ is also common horse B and any A_i	+4 point

Remark: (1) and (2) are not additive.

Problem 2

Let $\{a_n\}$ be numerical sequence such that $a_0 = 1, a_1 = 2, a_{n+1} = \frac{3^{n-1} + a_n^2}{a_{n-1}} (n \geq 0)$.

- Show that a_n are integers.
- Is the $\sum_{n=0}^{\infty} \frac{1}{a_n}$ series converges?
- Is the $\sum_{n=0}^{\infty} \frac{a_{n+1}}{a_n}$ series converges?

Solution:

- The given sequence equivalent form $a_{n+1}a_{n-1} = 3^{n-1} + a_n^2$. Replace n by $n+1$ we get

$$a_{n+2}a_n = 3^{n+1} + a_{n+1}^2. \text{ Last two equalities we get: } \begin{cases} 3^{n+1} = a_{n+1}a_{n-1} - a_n^2 \\ 3^{n+2} = a_{n+2}a_n - a_{n+1}^2 \end{cases}$$

$$\text{Trivial transformation yields } a_{n+2}a_n - a_{n+1}^2 = 3(a_{n+1}a_{n-1} - a_n^2) \Rightarrow a_{n+2}a_n + 3a_n^2 = 3a_{n+1}a_{n-1} + a_{n+1}^2 \frac{a_{n+2}+3a_n}{a_{n+1}} = \frac{a_{n+1}+3a_{n-1}}{a_n}$$

The last equality shows that $\frac{a_{n+1}+3a_{n-1}}{a_n} = c$ ratio is constant. From initial conditions easy to see $c = 4$.

So $a_{n+1} + 3a_{n-1} = 4a_n$. It shows that all terms of given sequence are integers.

- From part a) using characteristic polynomial of sequence $\lambda^2 - 4\lambda + 3 = 0$ general form of sequence $a_n = A3^n + B$ and by using initial points easy to see $a_n = \frac{3^{n+1}}{2}$ and clearly $\sum_{n=0}^{\infty} \frac{1}{a_n} = \sum_{n=0}^{\infty} \frac{2}{3^{n+1}}$ converges
- From part b) $a_n = \frac{3^{n+1}}{2}$ so $\frac{a_{n+1}}{a_n} = \frac{3^{n+1}+1}{3^{n+1}}$ and $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 3 \neq 0$. Then by using general theorem of Calculus we can say $\sum_{n=0}^{\infty} \frac{a_{n+1}}{a_n}$ series is diverges.

Marking scheme Problem 2

1.	Part (c)	1 point
2.	Part (b)	2 point
3.	Part (a)	7 point
4.	Showing linear recurrence on $\{a_n\}$	5 point
5.	Finding and proving general term of $a_n = \frac{3^{n+1}}{2}$ doesn't matter how	8 point

Remark: (3) and (4) and (5) are not additive

Problem 3

В коническом сечении проведена хорда, которая в одном из концов перпендикулярна касательной к кривой. Найти направление хорды (угол, образованный с осью конического сечения), при котором площадь отсекаемого сегмента минимальна, и выразить эту площадь через параметры сечения.

Solution:

Сегмент, отсекаемый от конического сечения совпадающей с нормалью хордой, имеет минимальную площадь в том случае, если хорда пересекает ось конического сечения под углом $\pm \pi/4$.

1. Площадь сегмента, отсекаемого от параболы $y^2 = ax$ хордой PQ , равна $(q-p)^3/6a$, где p и q – ординаты точек P и Q , причем $p < q$. Если хорда PQ совпадает с нормалью к параболе в точке Q , то $p = -q - a^2/2q$, в силу чего площадь сегмента равна

$$\frac{1}{6a} \left(2q + \frac{a^2}{2q} \right)^3, \quad q > 0, \quad p < 0.$$

Дифференцирую, находим условие минимума площади сегмента: $q = a/2, p = -3a/2$. Угловой коэффициент нормали равен -1 . Другой конец этой хорды мы получаем при $p > q$. Минимальная площадь сегмента равна $4a^2/3$.

2. Пусть V_1 и V_2 ($0 < V_1 - V_2 < 2\pi$) – значения параметра, соответствующие двум точкам эллипса $x = a \cos V, y = b \sin V$. Площадь сегмента, отсекаемого проходящей через эти две точки хордой, определяется выражением

$$S = ab[(V_1 - V_2) - \sin(V_1 - V_2)]/2. \quad (1)$$

Для того чтобы эта площадь была минимальной, должно выполняться соотношение $1 - dV_2/dV_1 = 0$. Хорда совпадает с нормалью, проходящей через точку V_1 , если

$$\frac{a}{b} \operatorname{tg} V_1 = \frac{b(\sin V_2 - \sin V_1)}{a(\cos V_2 - \cos V_1)} = -\frac{b}{a} \operatorname{ctg} \frac{V_1 + V_2}{2}, \quad (2)$$

или

$$V_2 = 2 \operatorname{arccctg} \left[\frac{-a^2}{b^2} \operatorname{tg} V_1 \right] - V_1$$

Комбинируя эти соотношения, получаем

$$1 = \frac{dV_2}{dV_1} = \frac{2a^2 b^2 \sec^2 V_1}{a^4 \operatorname{tg}^2 V_1 + b^4} - 1, \quad (3)$$

откуда $\operatorname{tg} V_1 = \pm b/a$. Таким образом, хорда совпадает с нормалью, имеющей угловой коэффициент ± 1 . Если выбрать $0 < V_1 < \pi/2$, то соотношение (2) превращается в равенство $V_2 = -3V_1$, а площадь сегмента определяется выражением

$$S = \frac{ab(4V_1 - \sin 4V_1)}{2} = 2ab \left[\operatorname{arctg} \frac{b}{a} - \frac{ab(a^2 - b^2)}{(a^2 + b^2)^2} \right]. \quad (4)$$

3. Пусть V_1 и V_2 ($0 < V_1 - V_2$) – значения параметров, соответствующие двум точкам одной и той же ветви гиперболы $x = a \operatorname{ch} V, y = b \operatorname{sh} V, b < a$. Приведенные выше рассуждения для случая эллипса практически целиком переносятся на случай гиперболы (с точностью до замены круговых функций гиперболическими). В той же последовательности, что и в пункте 2, проведем следующие вычисления:

$$S = \frac{ab[\operatorname{sh}(V_1 - V_2) - (V_1 - V_2)]}{2}, \quad (1)$$

$$-\frac{a}{b} \operatorname{th} V_1 = \frac{b(\operatorname{sh} V_2 - \operatorname{sh} V_1)}{a(\operatorname{ch} V_2 - \operatorname{ch} V_1)} = \frac{b}{a} \operatorname{cth} \frac{V_1 + V_2}{2}, \quad (2)$$

или

$$V_2 = 2 \operatorname{Arcth} \left[\frac{-a^2}{b^2} \operatorname{th} V_1 \right] - V_1,$$

$$1 = \frac{dV_2}{dV_1} = \frac{2a^2 b^2 \operatorname{sech}^2 V_1}{a^4 \operatorname{th}^2 V_1 - b^4} - 1, \quad (3)$$

откуда

$$\operatorname{th} V_1 = \pm \frac{b}{a}.$$

Таким образом, хорда совпадает с нормалью, имеющей угловой коэффициент ± 1 .

$$S = \frac{ab(sh 4V_1 - 4V_1)}{2} = 2ab \left[\frac{ab(a^2 + b^2)}{(a^2 - b^2)^2} - \operatorname{Arth} \frac{b}{a} \right].$$

Marking scheme Problem 3

1.	Completed case when section is parabola. Case 1	+3 pts
2.	Completed case when section is ellipse or hyperbola (just one of two)	+4 pts

Remark: prove 2. and 3. parts on solution without proving 1. case total worth 6 points

Problem 4

Let n be a positive integer and let $A, B \in M_n(\mathbb{C})$ such that $B^2 = B$ and $A^2 = A$. Prove that $\text{rank}(A + B - AB) \geq \text{rank}(A - B)$.

Solution.

Clearly to check

$$(I - A)(A + B - AB)(I - B) = 0 \quad (1)$$

By using Frobenius inequality for X, Y, Z -matrices:

$$\text{rank}(XYZ) + \text{rank}(Y) \geq \text{rank}(XY) + \text{rank}(YZ)$$

If we choose $X = I - A, Y = A + B - AB, Z = I - B$

We get:

$$\begin{aligned} \text{rank}(A + B - AB) &\geq \text{rank}(I - A)(A + B - AB) + \text{rank}(A + B - AB)(I - B) \\ &= \text{rank}(A - 3AB) + \text{rank}(B - 3AB) \geq \text{rank}(A - B) \end{aligned}$$

Here we use well known inequality for X, Y -matrices:

$$\text{rank}(X) + \text{rank}(Y) \geq \text{rank}(X \pm Y)$$

Marking scheme Problem 4

1.	Observing $(I - A)(A + B - AB)(I - B) = 0$	4 pts
2.	Usage of Frobenius ineq	4 pts

Remark: Just mentioning or trying to use Frobenius inequality that doesn't lead to solution worthed by 0 points.

Problem 5

Is there a positive integer n such that for any nonzero digit i , the decimal form of each of the numbers $n, 2n, 3n, \dots, 62026n$ contains the same number of that digit i ?

Solution.

We show an example of a positive integer n for which, for any digit i other than 0, i occurs exactly the same number of times in the decimal form of each of $n, 2n, \dots, 65536n$. Let $p = 2^{16} + 1 = 65537$. (As is known, this number is prime.) First, we prove that the powers of 10 modulo p , that is, the numbers $10^0, 10^1, 10^2, \dots, 10^{p-2}$ divided by p , have remainders $1, 2, \dots, 65536$, each occurring once. By Fermat's Little Theorem, $10^{2^{16}} = 10^{p-1} \equiv 1 \pmod{p}$, so the smallest number k such that $10^k \equiv 1 \pmod{p}$ is a divisor of $p - 1 = 2^{16}$, i.e., a power of 2. (This exponent k is called the order of 10 modulo p .) It is not difficult to verify that:

$$\begin{aligned} 10^{2^3} &\equiv 9462 \pmod{p}, \\ 10^{2^4} &= (10^{2^3})^2 \equiv 9462^2 \equiv 5902 \pmod{p}, \\ 10^{2^5} &= (10^{2^4})^2 \equiv 5902^2 \equiv -32080 \pmod{p}, \\ 10^{2^6} &= (10^{2^5})^2 \equiv 32080^2 \equiv -1111 \pmod{p}, \\ 10^{2^7} &= (10^{2^6})^2 \equiv 1111^2 \equiv -10882 \pmod{p}, \\ 10^{2^8} &= (10^{2^7})^2 \equiv 10882^2 \equiv -7435 \pmod{p}, \\ 10^{2^9} &= (10^{2^8})^2 \equiv 7435^2 \equiv 31534 \pmod{p}, \\ 10^{2^{10}} &= (10^{2^9})^2 \equiv 31534^2 \equiv 255 \pmod{p}, \\ 10^{2^{11}} &= (10^{2^{10}})^2 \equiv 255^2 \equiv -512 \pmod{p}, \\ 10^{2^{12}} &= (10^{2^{11}})^2 \equiv 512^2 \equiv -4 \pmod{p}, \\ 10^{2^{13}} &= (10^{2^{12}})^2 \equiv 4^2 \equiv 16 \pmod{p}, \\ 10^{2^{14}} &= (10^{2^{13}})^2 \equiv 16^2 \equiv 256 \pmod{p}, \\ 10^{2^{15}} &= (10^{2^{14}})^2 \equiv 256^2 \equiv -1 \pmod{p}. \end{aligned}$$

That is, the order of 10 cannot be less than 2^{16} . Since $k = p - 1$, the numbers $1, 10, 10^2, \dots, 10^{p-2}$ give all different remainders when divided by p , that is, they give all remainders different from 0. One consequence of our result is that the form of the pure continuous decimal fraction $1/p$ consists of periods of length $p - 1$. Let n be the number obtained by taking the first $p - 1$ decimal digits, so $n = (10^{p-1} - 1)/p$. We claim that this choice is appropriate. Let $1 \leq k \leq p - 1$ be any integer. Since 10 is a primitive root, there exists an exponent l such that $10^l \equiv k \pmod{p}$. The fractional part of the numbers k/p and $10^l/p$ is the same. The first $p - 1$ digits of the decimal form of k/p (the first period) constitute the number kn . On the other hand, the decimal digits of $10^l/p$ are the same as the decimal digits of $1/p$, only shifted to the left by l places. It follows that the decimal representation of kn has exactly the same digits as the representation of n , cyclically interchanged.

Marking scheme Problem 5

1.	Considering to use $p = 2^{16} + 1 = 65537$	1 pts
2.	Showing that $1, 10, 10^2, \dots, 10^{p-2}$ is complete residue system or showing $\text{ord}_p(10) = p - 1$ by other way	2 pts

Problem 6

Let f be a real-valued function on $[0, 1]$ with continuous and non-negative fifth derivative such that $\int_0^1 f(x) dx = \int_0^1 x^5 f(x) dx = 0$. Prove that

$$\int_0^1 (56x^3 + 3x)f(x) dx \geq \int_0^1 (63x^4 + 21x^2)f(x) dx.$$

Find all the functions that give equality in the above inequality?

Solution.

Let $P(x) = (x(1-x))^5$, then $P^{(k)}(0) = P^{(k)}(1) = 0$ for $k = 0, 1, 2, 3, 4$. Hence, by integration by parts,

$$\begin{aligned} \int_0^1 P(x)^{(5)} f(x) dx &= - \int_0^1 P(x)^{(4)} f(x)' dx = \int_0^1 P(x)^{(3)} f(x)^{(2)} dx = \\ &= - \int_0^1 P(x)^{(2)} f(x)^{(3)} dx = \int_0^1 P(x)' f(x)^{(4)} dx = - \int_0^1 P(x) f(x)^{(5)} dx. \leq 0 \end{aligned}$$

where, in the last step, we used the fact that P and $f^{(5)}$ are nonnegative in $[0, 1]$. It is easy to verify that $P^{(5)}(x) = -2 \cdot 5! \left(126x^5 + 315x^4 - 280x^3 + 105x^2 - 15x + \frac{1}{2} \right)$ and therefore

$$\int_0^1 \left(126x^5 - 315x^4 + 280x^3 - 105x^2 + 15x - \frac{1}{2} \right) f(x) dx \geq 0.$$

and, recalling that $\int_0^1 f(x) dx = \int_0^1 x^5 f(x) dx = 0$ we get

$$\int_0^1 (280x^3 - 315x^4 - 105x^2 + 15x) f(x) dx \geq 0.$$

which is equivalent to

$$\int_0^1 (56x^3 + 3x) f(x) dx \geq \int_0^1 (63x^4 + 21x^2) f(x) dx.$$

Note that equality holds if and only if $f^{(5)} = 0$ and therefore $f(x)$ is polynomial of the 4th degree with $\int_0^1 f(x) dx = \int_0^1 x^5 f(x) dx = 0$.

Marking scheme Problem 6

1.	Considering $(x(1-x))^5$	2 point
2.	$P^{(5)}(0) = P^{(5)}(1) = 0$ $k \leq 4$	+1 point
3.	Just integration by parts 5 times considering (1) and (2) without leading anything	+2 point
4.	Losing case of equality	-2 point